Circles

Class 2

Note for the teacher.

- **Aims.** Students locate the centre of a circle from two chords. They explain why the centres of all circles through two points lie on the perpendicular bisector of the segment, and why exactly one circle passes through three non-collinear points. They construct circumcircles and predict where the circumcentre lies. They explain why equal chords are equidistant from the centre, and the converse, and why a longer chord is nearer the centre.
- **NCERT links.** Theorem 5 is NCERT Theorem 1. Theorems 6, 7 and 8 keep their NCERT numbers. The questions include NCERT Exercise Sets 5.1, 5.4 and 5.5, the two *Think, Draw and Infer* questions, and End-of-Chapter Questions 12, 13, 17 and 23.
- **Prior work.** Class 1: Theorems 1 to 4, and the fact that the perpendicular bisector of every chord passes through the centre. Students use $\left(\frac{c}{2}\right)^2 + d^2 = r^2$ with confidence.
- **Materials.** Two A4 sheets, a bangle (or any round lid) and the geometry box for each student.
- **Method.** As in Class 1, students solve each *Solve first* set before the theorem is stated.

1 Finding the centre of a circle

Jamuna has drawn a circle by tracing around a bangle. She wants to find its centre. Her friend Amina says, "Fold the paper twice." Solve P1 to see why this works.

Solve first.

P1. Trace around a bangle on an A4 sheet. Mark two points A and B on the circle, not opposite each other, and join them with a ruler to draw the chord AB. Hold the sheet against a window and fold it so that A falls exactly on B. Press the fold firmly to make a crease, and open the sheet.

(a) The crease is the perpendicular bisector of AB. Which special point of the circle must it pass through? Use a fact from Class 1.

(b) Mark two more points C and D on the circle and draw the chord CD, so that CD is not parallel to AB. Fold again so that C falls exactly on D, and make a second crease. Where do the two creases meet?

(c) Why is one crease not enough to find the centre?

(d) What goes wrong if CD is parallel to AB?

Answers.

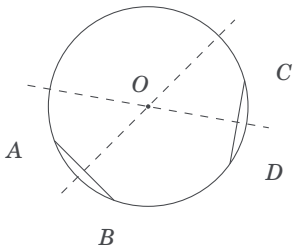
P1.

(a) The centre. The perpendicular bisector of every chord passes through the centre (Class 1).

(b) Both creases pass through the centre, and two lines that are not parallel meet in exactly one point. So the creases meet at the centre.

(c) One crease is a line. It contains the centre, but it contains infinitely many other points too.

(d) Both creases are perpendicular to the same direction and both pass through the centre. So they are the same line, and no single point is fixed.



Note for the teacher. Students may cut the circle out if scissors are available; folding the full A4 sheet against a window works equally well. Each crease is a diameter, and every diameter is a line of reflection symmetry of the circle. A circle also looks the same after any rotation about its centre: it has complete rotational symmetry.

Things to Remember

To find the centre of a circle, draw two chords that are not parallel. Their perpendicular bisectors meet at the centre.

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2 Circles through two points

Solve first.

P2. On an A4 sheet, draw a line segment $AB = 8$ cm and mark its midpoint M.

(a) Draw the circle with centre M and radius 4 cm. Does it pass through A and B? Can a circle with a smaller radius pass through A and B? Give a reason.

(b) Construct the perpendicular bisector of AB. With centre A and radius 5 cm, draw arcs that cut the bisector at O_1 above AB and at O_2 below AB. Draw the circle with centre O_1 and radius 5 cm, and the circle with centre O_2 and radius 5 cm. Do both circles pass through B? Explain why. Measure O_1M, and then find O_1M by calculation.

(c) How many circles of radius 5 cm pass through A and B?

(d) Try to draw a circle of radius 3 cm that passes through A and B. What goes wrong? Give a reason.

P3. Use P2 to answer these questions.

(a) Where do the centres of all the circles through A and B lie?

(b) As the centre moves away from M, does the radius increase or decrease?

(c) Is there a largest circle through A and B?

Answers.

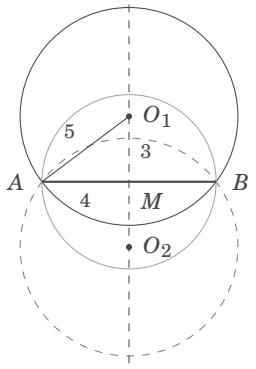
P2.

(a) Yes: $MA = MB = 4$ cm, so AB is a diameter of this circle. A smaller radius is impossible: AB would be a chord, and no chord is longer than the diameter.

(b) Yes. O_1 lies on the perpendicular bisector of AB, so $O_1B = O_1A = 5$ cm; the same holds for O_2. In right $\triangle O_1MA$, $O_1M^2 = 5^2 - 4^2 = 9$, so $O_1M = 3$ cm (measured: about 3 cm).

(c) Two: the centres O_1 and O_2, one 3 cm on each side of AB.

(d) An arc of radius 3 cm with centre A does not reach the perpendicular bisector, so no centre exists. The diameter would be 6 cm, but the chord AB would be 8 cm.



P3.

(a) On the perpendicular bisector of AB. Every point on it is equidistant from A and B, so every such point is the centre of a circle through A and B.

(b) The radius increases: $r^2 = OM^2 + 4^2$ grows as OM grows.

(c) No. The centre can move as far as we like, so the radius has no largest value. The circles become flatter near AB.

Things to Remember

Infinitely many circles pass through two points A and B. Their centres lie on the perpendicular bisector of AB. The smallest has AB as a diameter.

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3 Circles through three points

Solve first.

P4. On an A4 sheet, draw a straight line. Mark three points A, B and C on it, in this order, with $AB = 4$ cm and $BC = 6$ cm.

(a) Construct the perpendicular bisector of AB and the perpendicular bisector of BC.

(b) Do they meet? Give a reason.

(c) Can a circle pass through A, B and C?

P5. On an A4 sheet, draw any triangle ABC with sides between 5 cm and 10 cm.

(a) Construct the perpendicular bisectors of AB and BC. Mark their meeting point O.

(b) Measure OA, OB and OC.

(c) Explain, without measuring, why $OA = OB = OC$.

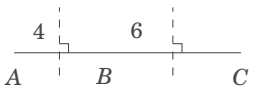
(d) Does the perpendicular bisector of CA also pass through O? Why?

Answers.

P4.

(b) No. Both bisectors are perpendicular to the same line ABC, so they are parallel.

(c) No. A centre would have to lie on both bisectors, and parallel lines have no common point.



P5.

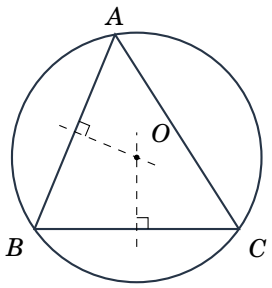
(b) The three lengths are equal (within about 1 mm).

(c) O is on the perpendicular bisector of AB, so $OA = OB$. O is on the perpendicular bisector of BC, so $OB = OC$. Hence $OA = OB = OC$.

(d) Yes. $OA = OC$, so O is equidistant from C and A and lies on the perpendicular bisector of CA.

Note for the teacher. P5(d) shows that the three perpendicular bisectors of a triangle always meet at one point. Students often draw all three and treat the third as a check of accuracy only.

Theorem 5 (NCERT Theorem 1). *There is a unique circle passing through three non-collinear points.*



Why it works.

Given: three points A, B and C that are not on one line.
To show: exactly one circle passes through A, B and C.

1. The centre of any circle through A and B lies on the perpendicular bisector of AB (P3). The centre of any circle through B and C lies on the perpendicular bisector of BC.
2. AB and BC are not on one line, so their perpendicular bisectors are not parallel. They meet in exactly one point O.
3. $OA = OB = OC$, so the circle with centre O and radius OA passes through all three points.
4. Any circle through the three points must have its centre at this point O and radius OA. So there is only one such circle.

This circle is the **circumcircle** of $\triangle ABC$. Its centre O is the **circumcentre**, and its radius is the **circumradius**.

Solve first.

P6. Construct each triangle with ruler, compass and protractor. Then construct its circumcircle: draw the perpendicular bisectors of two sides, mark their meeting point O, and draw the circle with centre O through one vertex. Say whether O lies inside the triangle, on a side, or outside the triangle.

(a) $AB = 5$ cm, $\angle A = 70°$, $\angle B = 60°$.

(b) $PQ = 6$ cm, $\angle P = 90°$, $PR = 8$ cm.

(c) $XY = 5$ cm, $\angle X = 100°$, $XZ = 4$ cm.

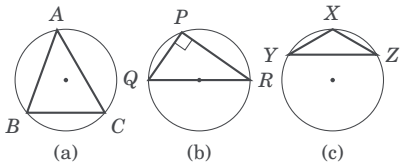
Answers.

P6.

(a) Acute-angled triangle ($\angle C = 50°$): the circumcentre is inside. Circumradius about 3.3 cm.

(b) Right-angled triangle: the circumcentre is the midpoint of the hypotenuse QR. $QR = 10$ cm, so the circumradius is 5 cm.

(c) Obtuse-angled triangle: the circumcentre is outside, beyond the longest side YZ. Circumradius about 3.5 cm.

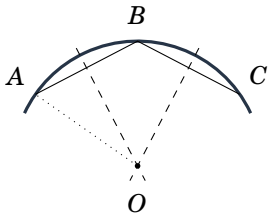


Note for the teacher. Give one triangle to each group and let the groups compare. Part (b) needs one more fact before it can be proved: the angle in a semicircle is 90°. That is done in Class 3.

Things to Remember

The circumcentre of a triangle lies inside an acute-angled triangle, at the midpoint of the hypotenuse of a right-angled triangle, and outside an obtuse-angled triangle.

Example 1. A potter finds a broken piece of a round clay plate. Explain how to find the centre and the radius of the full plate.



Given: a piece of the rim of a circle.
To find: the centre and the radius of the circle.

1. Place the piece on paper and trace its curved edge. Mark three points A, B and C on the traced arc.
2. Draw the chords AB and BC, and construct their perpendicular bisectors. They meet at O.
3. O is the centre: the perpendicular bisector of every chord passes through the centre, and only one circle passes through A, B and C (Theorem 5).

4. Measure OA. This is the radius.

Answer: the centre is the point where the two perpendicular bisectors meet, and the radius is OA.

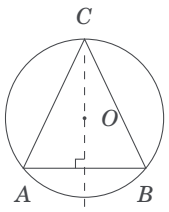
Exercise 2A

1. Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 7$ cm and $CA = 7$ cm. Construct its circumcircle, and let the circumcentre be O. Measure OA, OB and OC.

   Solution.

   Steps of construction:

   1. Draw $AB = 6$ cm.
   2. With A as centre and radius 7 cm, draw an arc.
   3. With B as centre and radius 7 cm, draw an arc cutting the first arc at C; join AC and BC, so $CA = CB = 7$ cm.
   4. Construct the perpendicular bisectors of AB and BC; they meet at O, the circumcentre.
   5. Draw the circle with centre O and radius OA.

   $\triangle ABC$ is isosceles and acute-angled, so O lies inside it, on the perpendicular bisector of AB (Theorem 5).
   Answer: $OA = OB = OC \approx 3.9$ cm.

2. Points A and B are 10 cm apart. What is the least possible radius of a circle through A and B? Is there a greatest possible radius? Explain.

   Solution.

   Given: $AB = 10$ cm.
   To find: the least possible radius of a circle through A and B; whether a greatest possible radius exists.

   1. The smallest circle has AB as a diameter, so its centre is the midpoint of AB (P2).
   2. Least radius $= \dfrac{AB}{2} = 5$ cm (radius $= \dfrac{\text{diameter}}{2}$).
   3. The centre of every circle through A and B lies on the perpendicular bisector of AB, and can be as far from AB as we like (P3), so the radius has no greatest value.

   Answer: least radius 5 cm; there is no greatest radius.

3. Points P and Q are 6 cm apart. A circle of radius 5 cm passes through P and Q.

   (a) Find the distance of its centre from the line PQ.

   (b) How many such circles are there?

   (c) Explain why no circle of radius 2 cm passes through P and Q.

   Solution.

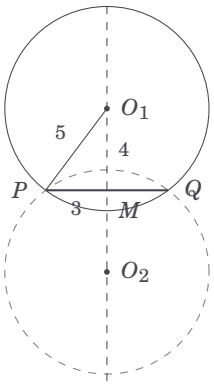
Given: $PQ = 6$ cm, radius $= 5$ cm, centre O; $OM \perp PQ$.
To find: (a) OM; (b) the number of such circles; (c) why no circle of radius 2 cm passes through P and Q.

(a) M is the midpoint of PQ, so $PM = 3$ cm (Theorem 4). In right $\triangle OMP$, $OM^2 = 5^2 - 3^2 = 16$, so $OM = 4$ cm (Pythagoras theorem).

(b) O lies on the perpendicular bisector of PQ, 4 cm from M; there is one such point on each side of PQ, so there are two circles.

(c) Its diameter would be 4 cm, but $PQ = 6$ cm would be a chord, and no chord is longer than the diameter.

Answer: (a) 4 cm; (b) two.



4. A, B and C lie on a straight line. Is there a point P with $PA = PB = PC$? Explain using the perpendicular bisectors of AB and BC. Can a straight line meet a circle in three points?

Solution.

No. P would lie on the perpendicular bisectors of AB and of BC, and both of these are perpendicular to the line ABC, so they are parallel and never meet. Hence there is no point equidistant from three collinear points, no circle passes through three collinear points, and a straight line meets a circle in at most two points.

5. Three villages lie at the corners of a triangle. The panchayat wants to build a water tank at the same distance from all three. Explain how to find the place on a map. What happens if the three villages lie on one straight road?

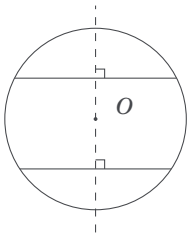
Solution.

Join the three villages to form a triangle and construct the perpendicular bisectors of two of its sides. The tank is built at their point of intersection, the circumcentre, because it is equidistant from all three villages (Theorem 5). If the villages lie on one straight road, the two bisectors are parallel and no such place exists (Question 4).

6. Kiran wants to find the centre of a circular plate drawn on paper. He draws two parallel chords and folds the sheet so that the ends of each chord meet. Find his mistake, and correct the method.

Solution.

Kiran is wrong. Both creases are perpendicular to the same pair of parallel chords and both pass through the centre, so they are the *same* line and do not fix a point (P1). He must draw two chords that are *not* parallel; their perpendicular bisectors then meet at the centre.



7. The circumcircle of $\triangle ABC$ is drawn. Can another triangle, congruent to $\triangle ABC$, have the same circumcircle? Explain.

Solution.

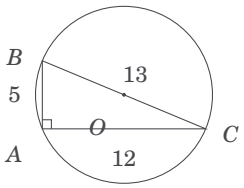
Yes, infinitely many. Turn $\triangle ABC$ about the circumcentre O through any angle: the image is congruent to $\triangle ABC$, and its vertices stay on the same circle because $OA = OB = OC$ (Theorem 5). Reflecting in a diameter also works.

8. The sides of a right-angled triangle are 5 cm, 12 cm and 13 cm. Where is its circumcentre? Find the circumradius.

Solution.

Given: a triangle with sides 5 cm, 12 cm and 13 cm.
To find: the position of the circumcentre and the circumradius.

1. $5^2 + 12^2 = 169 = 13^2$, so the triangle is right-angled and 13 cm is its hypotenuse (Pythagoras theorem).
2. The circumcentre of a right-angled triangle is the midpoint of the hypotenuse (P6).



Answer: circumcentre at the midpoint of the hypotenuse; circumradius $= \dfrac{13}{2} = 6.5$ cm.

9. The circumcentre of $\triangle ABC$ lies on the side BC. What kind of triangle is $\triangle ABC$? Draw a rough figure of the triangle and its circumcircle, with the centre O on BC, to support your answer.

Solution.

A right-angled triangle, with the right angle at A. In P6 the circumcentre lay on a side only for the right-angled triangle, and that side was the hypotenuse, so BC is the hypotenuse.

Note for the teacher. Question 9 is checked by construction here. Students prove it in Class 3, using the angle in a semicircle.

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4 Chords and their distance from the centre

Recall from Class 1: the distance of a chord from the centre is the length of the perpendicular from the centre to the chord.

Solve first.

P7. A circle has centre O and radius 5 cm. Chords AB and CD are each 8 cm long. $OM \perp AB$ and $ON \perp CD$.

(a) Find OM and ON.
(b) What do you notice?
(c) Would the answer to (b) change for two other equal chords of this circle? Why?

P8. In the same circle, chords PQ and RS are each 4 cm from O. Find PQ and RS.

P9. In the same circle, copy and complete the table.

Distance from O (cm)	0	3	4	5
Length of chord (cm)				

(a) As the chord moves away from the centre, what happens to its length?
(b) Which chord is the longest? Where does it lie?
(c) Of two unequal chords, which one is nearer the centre?

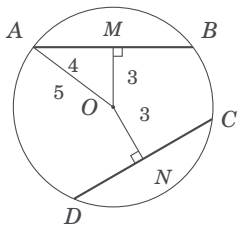
Answers.

P7.

(a) $AM = 4$ cm (the perpendicular from the centre bisects the chord, Theorem 4). In right $\triangle OMA$, $OM^2 = 5^2 - 4^2 = 9$, so $OM = 3$ cm. In the same way, $ON = 3$ cm.

(b) The equal chords are at equal distances from O.

(c) No. For any chord of length 8 cm the calculation uses the same radius and the same half chord, so the distance is always 3 cm.



P8. In right $\triangle OMP$, $PM^2 = 5^2 - 4^2 = 9$, so $PM = 3$ cm and $PQ = 6$ cm. In the same way, $RS = 6$ cm. Chords at equal distances from the centre are equal.

P9.

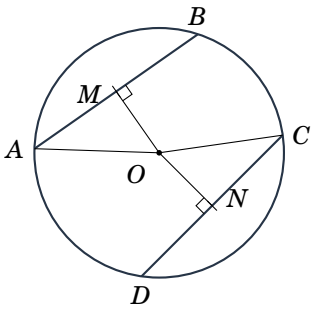
Distance from O (cm)	0	3	4	5
Length of chord (cm)	10	8	6	0

(a) It becomes shorter. At distance 5 cm the "chord" is a single point.

(b) The diameter, 10 cm. It passes through the centre (distance 0).

(c) The longer chord.

Theorem 6 (NCERT Theorem 6). *Chords of a circle having the same length are all at the same distance from the centre of the circle.*



Why it works.

Given: chords AB and CD of a circle with centre O, $AB = CD$, $OM \perp AB$ and $ON \perp CD$.
To show: $OM = ON$.

1. M and N are the midpoints of AB and CD (Theorem 4). Since $AB = CD$, $AM = CN$.
2. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $AM = CN$.
3. So $\triangle OMA \cong \triangle ONC$ (RHS), and $OM = ON$.

Theorem 7 (NCERT Theorem 7). *Chords of a circle that are equidistant from the centre have equal length.*

Why it works.

Given: chords AB and CD of a circle with centre O, $OM \perp AB$, $ON \perp CD$ and $OM = ON$ (same figure).
To show: $AB = CD$.

1. In $\triangle OMA$ and $\triangle ONC$: $\angle OMA = \angle ONC = 90°$, $OA = OC$ (radii) and $OM = ON$ (given).
2. So $\triangle OMA \cong \triangle ONC$ (RHS), and $AM = CN$.
3. $AB = 2AM$ and $CD = 2CN$ (Theorem 4), so $AB = CD$.

Theorem 8 (NCERT Theorem 8). *Let AB and CD be two chords of a circle with centre O, with $AB > CD$. Then the distance of AB from O is less than the distance of CD from O.*

Why it works.

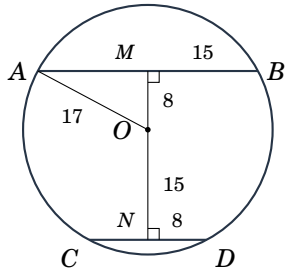
Given: $AB > CD$, $OM \perp AB$ and $ON \perp CD$ (same figure, with AB longer).
To show: $OM < ON$.

1. By Pythagoras theorem, $OM^2 + AM^2 = OA^2$ and $ON^2 + CN^2 = OC^2$. Since $OA = OC$, $OM^2 + AM^2 = ON^2 + CN^2$.
2. $AM = \dfrac{AB}{2}$ and $CN = \dfrac{CD}{2}$. Since $AB > CD$, $AM > CN$, so $AM^2 > CN^2$.
3. The two sums are equal, so $OM^2 < ON^2$, and $OM < ON$.

Things to Remember

For a chord at distance d from the centre of a circle of radius r, the length of the chord is $2\sqrt{r^2 - d^2}$. Equal chords are at equal distances from the centre, and the longer of two chords is nearer the centre. The diameter is the longest chord.

Example 2. A circle has centre O and radius 17 cm. Chord AB is 30 cm long and chord CD is 16 cm long. Which chord is nearer the centre? Find the distance of each chord from O.

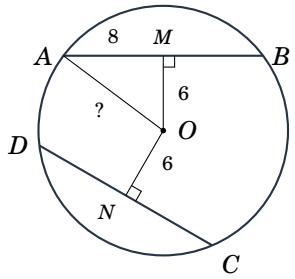


Given: $r = 17$ cm, $AB = 30$ cm, $CD = 16$ cm; $OM \perp AB$, $ON \perp CD$.
To find: which chord is nearer O; OM and ON.

1. $AB > CD$, so AB is nearer the centre (Theorem 8).
2. $AM = 15$ cm (Theorem 4). In right $\triangle OMA$, $OM^2 = 17^2 - 15^2 = 289 - 225 = 64$, so $OM = 8$ cm.
3. $CN = 8$ cm. In right $\triangle ONC$, $ON^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $ON = 15$ cm.

Answer: AB is nearer the centre; it is 8 cm from O, and CD is 15 cm from O.

Example 3. Chords AB and CD of a circle with centre O are each 6 cm from O, and $AB = 16$ cm. Find CD and the radius of the circle.



Given: $OM \perp AB$, $ON \perp CD$, $OM = ON = 6$ cm, $AB = 16$ cm.
To find: CD and the radius OA.

1. The chords are equidistant from the centre, so $CD = AB = 16$ cm (Theorem 7).
2. $AM = 8$ cm (Theorem 4). In right $\triangle OMA$, $OA^2 = 6^2 + 8^2 = 100$, so $OA = 10$ cm.

Answer: $CD = 16$ cm and the radius is 10 cm.

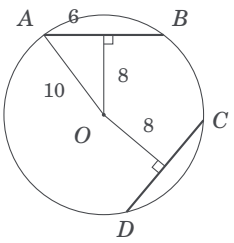
For every question in Exercise 2B, draw a rough figure: the circle, its centre O, the chords, and the perpendiculars from O to the chords. Mark all the given measurements on it.

Exercise 2B

1. A circle has centre O and radius 10 cm. Chords AB and CD are each 12 cm long. Find the distance of AB from O. Which theorem gives the distance of CD without a second calculation?

   Solution.

   Given: $r = 10$ cm, $AB = CD = 12$ cm; $OM \perp AB$.
   To find: the distance of AB from O, and of CD from O.

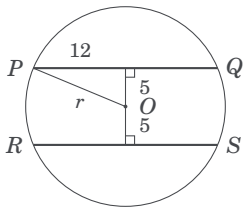
   1. $OM \perp AB$, so $AM = \dfrac{AB}{2} = 6$ cm (Theorem 4).
   2. In right $\triangle OMA$, $OM^2 = OA^2 - AM^2 = 10^2 - 6^2 = 64$, so $OM = 8$ cm (Pythagoras theorem).
   3. $CD = AB$, so CD is also 8 cm from O (Theorem 6).

   Answer: 8 cm each; Theorem 6.

2. Chords PQ and RS of a circle with centre O are each 5 cm from O, and $PQ = 24$ cm. Find RS and the radius.

   Solution.

Given: PQ and RS are chords of a circle with centre O, each 5 cm from O, and $PQ = 24$ cm.
To find: RS and the radius.



1. PQ and RS are equidistant from O, so $RS = PQ = 24$ cm (Theorem 7).
2. $PM = \dfrac{PQ}{2} = 12$ cm, where $OM \perp PQ$ (Theorem 4).
3. In right $\triangle OMP$, $OP^2 = OM^2 + PM^2 = 5^2 + 12^2 = 169$, so $OP = 13$ cm (Pythagoras theorem).

Answer: $RS = 24$ cm, radius 13 cm.

3. Use Pythagoras theorem to explain why Theorem 6 is true. Then use it to explain why Theorem 7 is true.

Solution.

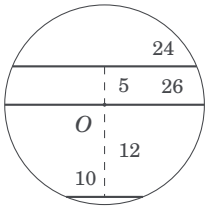
Given: chords AB and CD of a circle with centre O; $OM \perp AB$ and $ON \perp CD$.
To prove: (i) if $AB = CD$ then $OM = ON$; (ii) if $OM = ON$ then $AB = CD$.
Proof:

1. In right $\triangle OMA$, $OM^2 = OA^2 - AM^2$; in right $\triangle ONC$, $ON^2 = OC^2 - CN^2$ (Pythagoras theorem).
2. (i) $AB = CD$, so $AM = CN$ (Theorem 4). Also $OA = OC$ (radii). Hence $OM^2 = ON^2$, so $OM = ON$.
3. (ii) $OM = ON$, so $AM^2 = OA^2 - OM^2 = OC^2 - ON^2 = CN^2$, and $AM = CN$.
4. $AB = 2AM$ and $CD = 2CN$ (Theorem 4), so $AB = CD$.
5. Hence proved.

4. A circle has radius 13 cm. It has chords of lengths 10 cm, 24 cm and 26 cm. Without calculating, arrange them from the nearest to the farthest from the centre. Then find each distance to check.

Solution.

Given: $r = 13$ cm; chords of lengths 10 cm, 24 cm and 26 cm.
To find: the order of the chords from nearest to farthest from the centre, and each distance.



1. The longer chord is nearer the centre (Theorem 8), so the order is 26 cm, 24 cm, 10 cm.
2. 26 cm is a diameter, so its distance from the centre is 0 cm.
3. 24 cm: half is 12 cm, so $d^2 = 13^2 - 12^2 = 25$ and $d = 5$ cm (Pythagoras theorem).
4. 10 cm: half is 5 cm, so $d^2 = 13^2 - 5^2 = 144$ and $d = 12$ cm (Pythagoras theorem).

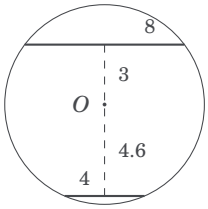
Answer: 0 cm, 5 cm, 12 cm.

5. In a circle of radius 5 cm, a chord of length 8 cm is 3 cm from the centre. Priya says, "A chord half as long, 4 cm, must be twice as far from the centre, 6 cm." Find her mistake.

Solution.

Priya is wrong. For the 4 cm chord, half the chord is 2 cm, so $d^2 = 5^2 - 2^2 = 21$ and $d = \sqrt{21} \approx 4.6$ cm, not 6 cm (Pythagoras theorem). Also no chord can be 6 cm from the centre, because 6 cm is more than the radius.

Distance and length are not in proportion; they are linked by $\left(\frac{c}{2}\right)^2 + d^2 = r^2$.



6. A chord of a circle of radius r is at distance d from the centre. Explain why the length of the chord is $2\sqrt{r^2 - d^2}$.

Solution.

Given: a chord AB of a circle with centre O and radius r, at distance d from O.
To prove: $AB = 2\sqrt{r^2 - d^2}$.
Construction: draw $OM \perp AB$, meeting AB at M.
Proof:

1. M is the midpoint of AB, so $AM = \dfrac{AB}{2}$ (Theorem 4).
2. $\triangle OMA$ is right-angled at M, so $AM^2 = OA^2 - OM^2 = r^2 - d^2$ (Pythagoras theorem).
3. So $AM = \sqrt{r^2 - d^2}$, and $AB = 2AM = 2\sqrt{r^2 - d^2}$.
4. Hence proved.

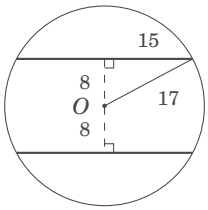
7. A circular park has radius 17 m and a fountain at its centre. Two straight footpaths cross the park, each 30 m long from edge to edge.

   (a) How far is each footpath from the fountain?

   (b) The two paths are parallel and on opposite sides of the fountain. Find the distance between them.

Solution.

Given: a circle of radius 17 m with centre O (the fountain); two parallel chords, each 30 m long, on opposite sides of O.
To find: (a) the distance of each path from the fountain; (b) the distance between the two paths.



(a) Each path is a chord of length 30 m, so half of it is 15 m (Theorem 4). Then $d^2 = 17^2 - 15^2 = 64$, so $d = 8$ m (Pythagoras theorem). The paths are equal chords, so they are at equal distances from O (Theorem 6); each is 8 m from the fountain.

(b) The perpendiculars from O to the two parallel paths lie on one line, and the paths are on opposite sides of O, so the distance between them is $8 + 8 = 16$ m.

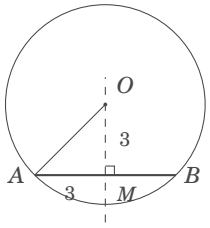
Answer: (a) 8 m each; (b) 16 m.

8. Draw a circle in which a chord of length 6 cm is at a distance of 3 cm from the centre. Write your steps, and measure the radius.

Solution.

Steps of construction:

1. Draw $AB = 6$ cm and construct its perpendicular bisector, meeting AB at M.
2. Mark O on the perpendicular bisector with $OM = 3$ cm.
3. Draw the circle with centre O and radius OA; then AB is a chord of length 6 cm and $OM \perp AB$ (Theorem 4).



$OA^2 = OM^2 + AM^2 = 3^2 + 3^2 = 18$, so $OA = \sqrt{18} \approx 4.2$ cm (Pythagoras theorem).

Answer: radius ≈ 4.2 cm.

9. True or false? Give a reason for each.

   (a) Two chords at the same distance from the centres of two different circles are equal.

   (b) Of two chords of a circle, the longer one is nearer the centre.

   (c) The chord farthest from the centre is the diameter.

Solution.

(a) False. Theorems 6 and 7 are about one circle, not two. Two chords 8 cm from the centres of circles of radii 10 cm and 17 cm have lengths 12 cm and 30 cm (Pythagoras theorem).

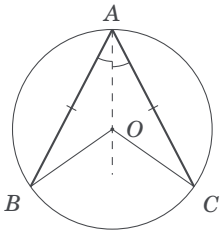
(b) True. In one circle, the longer of two chords is nearer the centre (Theorem 8).

(c) False. The diameter passes through the centre, so it is the *nearest* chord, at distance 0 (Theorem 8).

10. In a circle with centre O, chords AB and AC are equal. Explain why the centre O lies on the bisector of $\angle BAC$.

Solution.

Given: chords AB and AC of a circle with centre O, with $AB = AC$.
To prove: O lies on the bisector of $\angle BAC$.
Proof:



1. In $\triangle OAB$ and $\triangle OAC$: $OB = OC$ (radii), $AB = AC$ (given) and OA is common.
2. So $\triangle OAB \cong \triangle OAC$ (SSS), and $\angle OAB = \angle OAC$.
3. So AO bisects $\angle BAC$, and O lies on the bisector.
4. Or: $AB = AC$, so the chords are equidistant from O (Theorem 6), and a point equidistant from the two arms of an angle lies on its bisector.
5. Hence proved.

Note for the teacher. Questions 1 to 4 give the essential practice with Theorems 6 to 8. Question 5 must be discussed in class. If time is short, set Questions 7, 8 and 10 as homework.

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5 Further questions

F1. Find the length of a chord of a circle of radius 7 cm that is 6 cm from the centre. Leave the answer in square-root form.

Solution.

Given: $r = 7$ cm; the chord is 6 cm from the centre.
To find: the length of the chord.

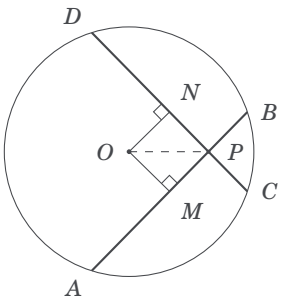
1. The perpendicular from the centre bisects the chord, so half the chord = $\sqrt{r^2 - d^2}$ (Theorem 4).
2. Half the chord = $\sqrt{7^2 - 6^2} = \sqrt{13}$ cm (Pythagoras theorem).
3. Length of the chord = $2\sqrt{13}$ cm.

Answer: $2\sqrt{13}$ cm, about 7.2 cm.

F2. Two equal chords AB and CD of a circle intersect at P. Show that $AP = CP$ and $PB = PD$.

Solution.

Given: equal chords AB and CD of a circle with centre O, intersecting at P.
To prove: $AP = CP$ and $PB = PD$.
Construction: draw $OM \perp AB$ and $ON \perp CD$.
Proof:

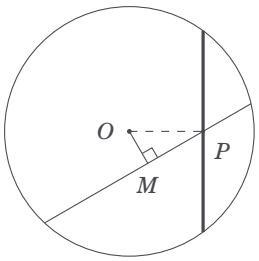


1. M and N are the midpoints of AB and CD, so $AM = CN$ (Theorem 4).
2. $AB = CD$, so $OM = ON$ (Theorem 6).
3. In $\triangle OMP$ and $\triangle ONP$: $\angle OMP = \angle ONP = 90°$, OP is common and $OM = ON$.
4. So $\triangle OMP \cong \triangle ONP$ (RHS), and $MP = NP$.
5. $AP = AM + MP = CN + NP = CP$ (from the figure).
6. $PB = AB - AP = CD - CP = PD$.
7. Hence proved.

F3. P is a point inside a circle with centre O (P is not O). Show that, of all the chords through P, the shortest one is perpendicular to OP.

Solution.

Given: a circle with centre O and a point P inside it, $P \neq O$; AB is a chord through P and $OM \perp AB$.
To prove: of all the chords through P, the shortest is perpendicular to OP.
Proof:



1. In right $\triangle OMP$, $OM \leq OP$, with equality only when $M = P$ (Pythagoras theorem).
2. When $M = P$, $AB \perp OP$, and the distance of AB from O is OP, the greatest possible.
3. A chord farther from the centre is shorter (Theorem 8), so the chord perpendicular to OP is the shortest of all chords through P.
4. Hence proved.

Note for the teacher. F3 leads into Class 3, which turns from chords to the angles that arcs subtend at the centre and on the circle.